

WAUKEGAN NATIONAL, IL, USA

WMO: 725347

Lat: 42.417N

Lon: 87.867W

Elev: 222

StdP: 98.69

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 14880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.6	-24.5	0.4	-18.5	-22.2	0.5	-15.9	11.3	2.1	10.3	0.5	3.2	270	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	31.9	23.4	30.0	22.6	28.3	21.7	24.7	29.9	23.7	28.2	22.8	27.0	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.2	27.6	22.3	17.4	26.6	21.4	16.5	25.5	76.0	29.8	71.9	28.3	68.5	27.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.8	DB	-22.6	34.4	3.3	2.0	-24.9	35.8	-26.8	37.0	-28.7	38.2	-31.1	39.6	(4)
(5)			WB	-23.0	26.3	3.2	0.9	-25.2	26.9	-27.1	27.4	-28.9	27.9	-31.2	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.8	-3.7	1.9	7.8	13.5	19.0	22.0	21.4	17.6	11.0	4.4	-2.0	(6)
(7)		DBStd	10.68	6.38	6.13	6.00	5.02	4.93	4.11	3.51	3.04	4.24	4.82	5.35	5.95	(7)
(8)		HDD10.0	1825	459	384	262	99	19	0	0	0	1	47	179	373	(8)
(9)		HDD18.3	3728	717	617	510	318	167	41	8	7	64	232	417	630	(9)
(10)		CDD10.0	1486	0	1	11	35	127	269	371	352	229	78	12	1	(10)
(11)		CDD18.3	348	0	0	1	3	17	60	121	101	42	5	0	0	(11)
(12)		CDH23.3	2733	0	0	5	41	151	527	1000	687	291	32	0	0	(12)
(13)		CDH26.7	785	0	0	1	8	32	160	323	187	71	4	0	0	(13)
(14)	Wind	WSAvg	3.6	4.0	4.0	4.0	4.3	3.7	3.1	2.8	2.7	3.0	3.6	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	913	48	41	57	96	101	105	90	99	81	75	69	51	(15)
(16)		PrecMax	1130	94	118	112	193	213	247	184	288	242	175	149	114	(16)
(17)		PrecMin	599	12	5	17	43	15	25	34	49	17	19	10	20	(17)
(18)		PrecStd	123	22	25	29	41	46	55	40	48	44	45	37	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.9	22.9	27.6	29.7	33.0	35.2	33.4	31.1	27.2	20.3	15.0	(19)
(20)			MCWB	10.3	11.1	16.1	17.5	20.8	23.2	24.6	23.9	22.6	20.2	14.8	12.7	(20)
(21)		2%	DB	8.8	11.2	17.5	23.6	27.2	30.9	32.1	30.9	28.7	23.5	17.3	11.1	(21)
(22)			MCWB	6.1	8.0	12.8	15.4	19.4	22.2	24.0	23.4	21.9	17.8	12.4	9.3	(22)
(23)		5%	DB	5.6	7.1	14.1	20.2	24.8	28.2	30.0	28.6	27.0	21.3	15.0	7.9	(23)
(24)			MCWB	3.4	4.3	10.0	14.0	18.3	21.5	23.1	22.2	20.1	16.6	11.3	6.0	(24)
(25)		10%	DB	3.4	4.0	10.6	16.8	22.4	26.3	28.2	27.1	24.5	18.8	12.5	6.0	(25)
(26)			MCWB	2.1	2.2	7.3	11.3	16.8	20.0	22.1	21.5	19.5	14.8	9.2	4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	12.6	17.0	19.0	22.6	25.2	26.3	25.5	24.1	21.6	15.7	12.8	(27)
(28)			MCDB	12.5	14.5	22.5	25.7	28.0	30.3	31.8	31.3	29.1	25.6	18.3	15.0	(28)
(29)		2%	WB	6.5	8.1	14.0	16.5	20.5	23.4	25.1	24.1	22.8	19.3	13.7	9.0	(29)
(30)			MCDB	8.5	11.0	17.2	22.0	25.4	28.7	30.5	28.6	26.8	21.9	16.6	10.4	(30)
(31)		5%	WB	3.6	4.5	10.6	14.5	19.1	22.2	24.0	23.2	21.6	17.3	11.6	6.5	(31)
(32)			MCDB	5.1	6.7	13.3	19.7	23.4	27.3	28.7	27.1	25.2	20.2	14.2	7.8	(32)
(33)		10%	WB	2.1	2.3	7.2	11.7	17.6	21.0	23.0	22.4	20.4	15.5	9.5	4.2	(33)
(34)			MCDB	3.5	4.0	10.2	16.5	21.5	25.4	27.1	25.8	23.6	18.4	12.4	5.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.6	9.9	10.3	10.1	9.6	9.2	9.7	9.0	8.2	7.2	(35)
(36)			MCDBR	9.7	10.8	13.5	15.9	13.7	12.3	11.2	11.1	11.7	12.1	11.1	8.9	(36)
(37)			MCWBR	8.0	8.2	9.4	9.7	7.7	6.3	5.2	5.1	5.9	7.4	7.9	7.7	(37)
(38)		5% WB	MCDBR	8.7	10.4	12.4	14.5	12.5	11.3	10.2	9.4	10.1	10.6	10.0	8.3	(38)
(39)			MCWBR	7.6	8.3	9.6	9.8	7.7	6.5	5.6	5.0	5.7	7.6	8.2	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.308	0.341	0.394	0.419	0.436	0.436	0.429	0.400	0.350	0.311	0.291	(40)	
(41)		taud	2.446	2.401	2.400	2.280	2.261	2.263	2.262	2.281	2.334	2.473	2.523	2.485	(41)	
(42)		Ebn at Noon	856	896	904	874	857	841	837	830	829	837	829	826	(42)	
(43)		Edh at Noon	75	92	105	128	134	134	133	126	111	86	70	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.49	2.37	3.55	4.60	5.28	6.05	6.13	5.31	4.33	2.80	1.75	1.23	(44)	
(45)		RadStd	0.19	0.29	0.29	0.38	0.41	0.40	0.28	0.33	0.37	0.32	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page